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Radiation Diversity Enabled Self-Isolated Compact Dual-Band Cubic MIMO Antenna for Wireless Biomedical Implants in Variable and Dynamic Environment

TAHIR BASHIR ¹, WEI LI ¹ (Member, IEEE), AND TIAN XIA ² (Senior Member, IEEE)

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¹College of Integrated Circuit Science and Engineering, Nanjing University of Posts and Telecommunications, Nanjing 210023, China²Department of Electrical and Biomedical Engineering, University of Vermont, Burlington, VT 05405 USA

CORRESPONDING AUTHOR: Wei Li (e-mail: wei.li@njupt.edu.cn).

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ABSTRACT This study presents a compact dual-band cubic multi-input-multi-output (MIMO) antenna specifically designed for gastrointestinal (GI) tract capsule endoscopy and cardiac leadless pacemaker systems. The proposed cubic MIMO antenna operates across two frequency bands: 1.395 to 1.4 GHz and 2.4 to 2.4835 GHz. Comprising four individual antennas, it has overall dimensions of $5.12 \times 5.12 \times 4.6$ mm³, which makes it a compact cubic design, achieved by employing symmetrically embedded radiating patch slots. The strategic relocation of port, pin, and ground slot not only resulted in reduced coupling due to opposite current flow but also contributed to achieving excellent frequency tuning for all antenna elements in cubic configuration. Encapsulated within wireless implants with batteries, sensors, and device circuitry, the proposed MIMO antenna was simulated in both homogeneous and heterogeneous body phantoms, including the small intestine, large intestine, stomach, and heart. Experimental validation also conducted using minced pork yielded results that agree with simulations, demonstrating the MIMO antenna effective performance, including measured reflection coefficient (−22 dB, −19 dB), gain (−28.17 dBi, −18.15 dBi), −10 dB bandwidth (390 MHz, 670 MHz), minimal coupling (−23 dB, −24 dB), and fractional bandwidth (27%, 26%) at 1.3975 and 2.45 GHz, respectively. Each cubic element radiates in four opposite directions, enabling radiation diversity in all four directions, crucial for various body postures during movement. The specific absorption rate (SAR) is also calculated and confirmed to remain within very safe limits for human implantation. Furthermore, a communication link analysis established the reliability of the antenna in maintaining stable communication with an external device over an 10 m and 15 m radius at the respective resonant frequencies, achieving a high data transmission rate of 100 Mbps. Further evaluation, including envelope correlation coefficient (ECC), diversity gain (DG), channel capacity loss (CCL), and total active reflection coefficient (TARC), confirms the usefulness of the proposed MIMO. Consequently, this MIMO antenna emerges as a highly promising candidate with radiation diversity, high compactness, and self-isolation ability for several wireless biomedical implants.

INDEX TERMS Implantable antenna, dual-band, MIMO, capsule endoscopy, leadless pacemaker, coupling, radiation diversity.

I. INTRODUCTION

The rapid growth of the global population, coupled with a significant increase in the proportion of elderly individuals, has imposed substantial burden on conventional healthcare systems. This population transformation has contributed to increased healthcare cost and a reduction in overall quality of care and patient well-being [1]. To effectively mitigate these challenges, future healthcare systems must prioritize scalable, cost-effective, and adaptable solutions. One promising approach involves the development and deployment of wireless implantable biomedical devices (WIMBDs) and wireless body area networks (WBANs), which function in close proximity to the human body [2]. These technologies enable continuous, real-time monitoring of vital biological and physiological parameters, thereby enhancing diagnostic precision, treatment efficacy, and long-term patient management. In recent years, the demand of WIMBDs has significantly increased due to their applications in health monitoring systems, including wireless capsule endoscopy (WCE), cardiac pacemakers, intracranial pressure monitors, glucose sensors, retinal prostheses, and head implants [3], [4], [5], [6], and [7]. These devices are crucial for real-time monitoring and data transmission to external devices and cloud platforms, assisting in accurate diagnosis. WIMBDs comprise multiple integrated components, including batteries, cameras, sensors, and antennas, all encapsulated within a biocompatible, capsule-shaped housing. Among these, the antenna is a critical element, as it facilitates wireless communication plays a pivotal role in ensuring device performance. However, designing a compact, high-data-rate antenna remains challenging due to severe space constraints and the complex in-body environment. Recent designs have employed single-input-single-output (SISO) topology, which offers ease of implementation but has several limitations for modern WIMBDs.

MIMO technology offers a promising solution for modern WIMBDs by enhancing communication reliability, channel capacity, and signal quality making it essential for real-time monitoring and diagnosis. Recent research has proposed various MIMO antenna designs to meet these requirements. A two-element MIMO antenna operating at 915 MHz incorporates reactive slots for compactness and reduced mutual coupling, delivering satisfactory performance [8]. Another two-element design at 2.45 GHz, sized $5.35 \times 6.2 \times 0.12 \text{ mm}^3$, achieves 28 dB isolation and a peak gain of -20.5 dBi [9], while a 402 MHz two-element MIMO antenna, though effective, is impractically large at $22 \times 22 \times 0.635 \text{ mm}^3$ [10]. A four-element antenna at 2450 MHz uses metamaterials to mitigate mutual coupling [11], and a quad-element MIMO antenna for 433 MHz improves data transmission and reduces multipath fading, though it suffers from frequency detuning [12]. Another four-element MIMO antenna employed an electromagnetic band-gap (EBG) structure faces low isolation issues [11]. Another four-port MIMO antenna for WCE, operating at 2.45 GHz with dimensions of $6.2 \times 5.2 \times 0.127 \text{ mm}^3$, demonstrates good performance [13]. Despite incorporating multiple elements, these designs are

restricted to single-frequency operation, which does not meet the multi-frequency needs of modern WIMBDs. Such devices must support various frequencies for communication protocols like wireless power transfer, data transmission, and remote control [14], [15], and [16]. To overcome single-band limitations, several multiband MIMO implantable antennas have also been developed in recent studies [5], [17], [18], [19], [20], [21], and [22]. The proposed planar configuration implantable MIMO antennas enhance communication performance, such as channel capacity, signal quality, and data transmission, they have inherent limitations. Their two-dimensional geometry restricts signal coverage from various angles, and in dynamic environments like capsule endoscopy, where the device rotates unpredictably, particularly in the intestines, planar antennas may fail to maintain reliable signals. Additionally, their flat structure limits radiation diversity, crucial for overcoming signal blockages caused by body tissues and posture changes. Additionally, single and dual-element omnidirectional circularly polarized (CP) antennas are effective and offer certain advantages such as compactness, easy to design, fabricate, polarization insensitivity, and improved robustness through limited spatial diversity they are may not ideally suited for highly variable and dynamic environments. Furthermore, single-antenna systems lack the capacity to support the high data rates required by next-generation implantable devices equipped with high-resolution cameras for real-time image and video transmission. Single CP antennas are valued for their simplicity and suitability in low-data-rate applications, while two-element CP MIMO configurations provide modest enhancements in link reliability. However, both designs may face limitations in maintaining highly stable communication within the complicated GI tract, where continuous movement, random orientation, and varying dielectric properties prevail.

In contrast, cubic configuration effective to ensure efficient communication in wireless health monitoring systems, such as capsule endoscopy and leadless pacemaker applications. As illustrated in Fig. 1, the patient is continuously moving and changing body posture. Despite these movements and the changing orientations of capsule devices within the GI tract of the patient, the cubic MIMO system ensures that at least one of the four antennas maintains reliable communication between the implantable device and the receiver. A good channel capacity, high data transmission rate and stable communication from multiple directions can be achieved by employing this configuration, thanks to its radiation diversity characteristics. The cubic MIMO offers several advantages as mentioned above in comparison to planar configuration, however it also has some limitations. To design a multi-band compact cubic MIMO antenna for modern WIMBDs with sufficient isolation, channel capacity, pattern diversity, and fit them into compact implantable devices for diverse biomedical applications can be challenging. The proposed work addresses all these challenges. The key contributions lies in the development of a highly efficient, compact, dual-band cubic MIMO antenna designed for several wireless biomedical implants,

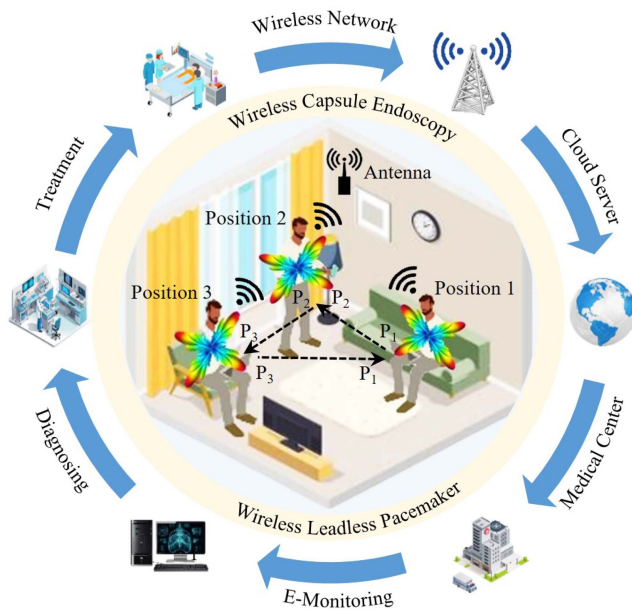


FIGURE 1. Overview of wireless health monitoring system using an encapsulated radiation diversified cubic MIMO implantable antenna for capsule endoscopy and cardiac leadless pacemaker systems in continuous movement, variable orientation of device within patient, and various body posture scenario.

including GI tract capsule endoscopy and cardiac leadless pacemaker systems for continuous movable, and different body posture scenarios, as shown in Fig. 1. In capsule endoscopy, the antenna is embedded in a compact, swallowable capsule that traverses the digestive tract to diagnose disorders by capturing and wirelessly transmitting images and physiological data to external devices. In cardiac leadless pacemakers, the antenna is integrated into a self-contained device implanted in the heart enable early diagnosis support timely clinical decisions. In both applications, the antenna must be compact and ensure reliable wireless communication within the complex and variable dielectric environment of internal organs. Designing an implantable antenna, there is always a tradeoff between the operating frequency and its overall performance. For instance, lower frequency bands, such as the Medical Implant Communications Service (MICS) band (402–405 MHz) and the Medical Device Radio Communications Service (MedRadio) band (401–406, 413–419, 426–432, 438–444, and 451–457 MHz), offer the advantage of lower power consumption, as signals experience less attenuation when passing through biological tissues. In contrast, higher frequency bands like the Wireless Medical Telemetry Service (WMTS) band (1.395–1.432 GHz) and the Industrial, Scientific, and Medical (ISM) bands (433.05–434.79 MHz, 2.400–2.480 GHz, and 5.725–5.875 GHz) allow for smaller antenna sizes, broader bandwidths, and faster data transmission rates. Therefore, selecting an appropriate frequency band for implantable antennas is crucial, particularly when considering the high loss and heterogeneous nature of human tissue. To achieve high data transmission rate, antenna

miniaturization as well as multiple communication protocols, the proposed MIMO works for both WMTS 1.4 GHz frequency band and ISM 2.45 GHz frequency bands for wireless power transfer and data telemetry applications. The proposed cubic MIMO occupies a minimum volume of 120.58 mm³, comprising four individual elements, and achieved good enough isolation between them without employing any external components while satisfying good MIMO antenna performances as well, including reflection coefficient, gain, isolation, bandwidth, and fractional bandwidth. To show the usefulness of the proposed antenna for several biomedical applications, we encapsulated the proposed antenna in several implantable devices (IMDs) and simulated in both homogeneous and heterogeneous environments. The proposed antenna, fabricated and encapsulated in realistic IMDs, and measurements were taken in minced pork and compared with simulated results, showing a good agreement with each other. In order to show how safe the presented MIMO antenna is for the human body, a SAR value has been calculated. Furthermore, a communication link analysis was also carried out to show the reliability of communication between the proposed MIMO and external devices for real-time data transmission in a radius of several meters with high data transmission rates. Furthermore, MIMO channel parameters, including ECC, DG, CCL, and TARC, have also been performed to satisfy the MIMO performance. Table 1 provides a detailed comparison of the proposed cubic MIMO antenna with recently reported implantable MIMO designs, highlighting its significant advantages for biomedical applications.

- *Miniaturization with Design Simplicity:* The proposed antenna achieves miniaturization through a simple design that incorporates a single ground slot and symmetrically embedded radiating patch slots. Unlike conventional meandered structures, which can be complex, the simple design makes the antenna easy to both design and fabricate, particularly when integrated with recently developed MIMO systems.
- *Self-Isolation of MIMO Elements:* Another key point of the proposed design is its ability to self-isolate closely spaced MIMO elements without relying on external components or large inter-element gaps. This feature enhances the efficiency and compactness of the antenna, making it suitable for space-constrained biomedical implants.
- *Alignment-Free Communication:* The antenna eliminates the need for precise alignment between the transmitter (patient) and receiver, ensuring stable communication within several meters of range. This capability is crucial for accommodating the natural movements and varying postures of patients, thus improving the practicality of the system in real applications.
- *Multi-Tissue cubic MIMO Antenna:* The proposed MIMO antenna is useful for various body tissues, ensuring stable performance. It is particularly well-suited for several in-body applications, including the small intestine, large intestine, stomach, and heart.

TABLE 1. Comparison of the Proposed Cubic MIMO Implantable Antenna With Very Recently Reported MIMO Implantable Antennas

Reference	[18]	[13]	[23]	[24]	[25]	[26]	[27]	This work
Year	2023	2023	2024	2024	2024	2024	2024	2025
Frequency [GHz]	0.915, 2.45	2.45	0.915	0.915, 1.78, 2.45	0.915, 2.45	0.61, 1.65, 2.45	2.45	1.3975, 2.45
Bandwidth	82.5, 148.5	200	900	90, 129, 200	130, 300	200, 200	500	390, 670
Isolation [dB]	-24	-20	-20	-25	-31, and -29	-23.1, -14.8, and -20.4	-17	-23 and -24
Gain [dBi]	-32.15, -22.2	-20.1	-30.4	-37.3, -30.3, -27.9	-30.47, -24.71	-28.9, -38.4, -41.7	-19.3	-28.17, -18.15
Implant Depth [mm]	58	50	65	60	32.5	60	25	50
Number of Elements	2	4	2	4	2	2	2	4
ECC	> 0.2	<0.12	< 0.1	< 0.26	0.003, 0.104	0.5	0.1	> 0.15 and > 0.1
DG [dB]	9.76	9.95	9.9	10	9.99, 9.95	9.8	9.9	< 9.2 and < 9.7
Decoupling Mechanism	No	No	No	No	Yes	No	No	Yes
Pattern diversity	No	No	No	No	No	No	No	Yes
Integrated Devices	Deep Tissue	Capsule	Capsule	Capsule	Cortical Visual Prostheses	Pacemaker	Capsule	Capsule, Pacemaker
Implant Depth Analysis	Yes	No	No	Yes	No	No	No	Yes
Front to Back Ratio [dB]	No	No	No	No	No	No	No	2.40, 1.30
Device Components Effect	Yes	Yes	No	No	No	No	No	Yes
Equivalent Circuit Model Analysis	No	No	No	No	No	No	Yes	Yes
Antenna profile	Planar	Planar	Conformal	Planar	Planar	Planar	Planar	Cubic
Antenna dimensions or volume [mm ³]	10.8×5.6×0.254	6.2×5.2×0.127	30.5×15×0.04	10.5×7.5×0.127	7×7×0.3	$\pi \times 4.5^2 \times 0.508$	D = 10, H = 4	5.12×5.12×4.6

D: Diameter of the antenna, H: Height of the antenna

- *Enhanced Performance for Biomedical Devices:* The antenna also stands out for its excellent overall performance, both in terms of individual antenna characteristics and MIMO functionality. It ensures a stable, high-speed communication link in all directions, making it ideal for use in implantable biomedical devices.

These attributes contribute to advancing the effectiveness of remote monitoring and diagnostic systems for patients. In addition to the usual simulation investigations conducted for implantable MIMO antennas such as scattering parameters, radiation patterns, SAR analysis, link budget, and MIMO channel parameters, we present a comprehensive and extensive simulation study that includes all of the above mentioned analysis, with a particular focus on aspects that have not been previously explored in the context of cubic MIMO configurations. Our work also investigates (1) influence of the cubic antenna structure on antenna tuning and detuning, (2) decoupling mechanism between closely connected antenna elements in cubic configuration, (3) gain deviation, (4) equivalent circuit model analysis of cubic MIMO, (5) front-to-back ratio (FBR), (6) impact of implant depth on antenna gain

in cubic configurations, and the (7) effect of device components on overall cubic MIMO antenna performance. These additional investigations provide new insights of cubic MIMO antennas for implantable biomedical devices.

The paper is organized as follows: Section II covers antenna and device design, decoupling, detuning, and equivalent circuit analysis. Section III presents multi-tissue simulations, fabrication, measurements, SAR, and communication analysis. Section IV discusses MIMO channel parameters. Section V concludes the study.

II. DESIGN METHODOLOGY OF PROPOSED CUBIC MIMO ANTENNA

In the development of a compact dual-band cubic MIMO implantable antenna with optimized performance for various biomedical applications, a comprehensive design process is considered. This process involves multiple phases: the initial design of implantable devices (such as capsules and leadless pacemakers) to ensure realistic results, followed by the design and tuning of a single antenna element at the targeted frequency bands. Subsequently, a cubic MIMO antenna is designed using a previously designed single element, with

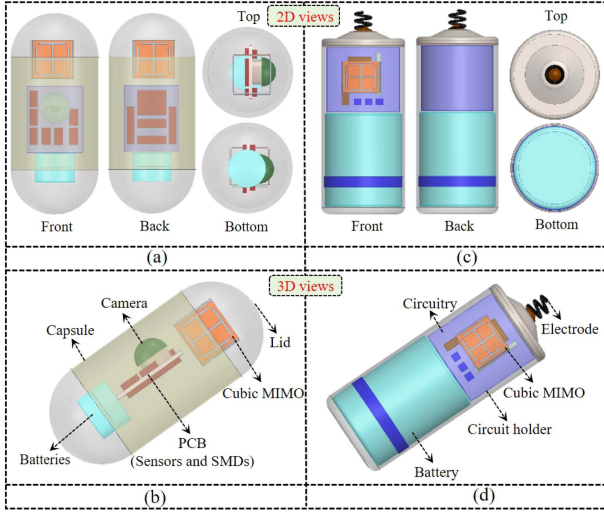


FIGURE 2. The detailed layout of designed WIMBDs: (a), (b) different views of GI tract capsule device; (c), (d) different views of cardiac leadless pacemaker device.

particular attention given to minimizing coupling between adjacent elements. Fabrication and measurements are then conducted, and the results are compared with simulations. Each phase of this design process is detailed in the following subsections.

A. CAPSULE-LIKE IMPLANTABLE DEVICES LAYOUT

This research emphasizes the development of a compact dual-band cubic MIMO antenna designed for several biomedical applications. To ensure proper insulation from surrounding biological tissues, biocompatible coatings are typically applied to implantable antennas. However, for practical deployment and system-level analysis, these antennas must be integrated into fully encapsulated devices. In this work, we focus on capsule-type devices to evaluate the antenna performance adaptability across a variety of simulation environments. Fig. 2 illustrates the devices layout from multiple perspectives. Fig. 2(a) presents various perspectives of the air-filled capsule endoscopy device, which has overall dimensions of 24.65 mm in length and 5.5 mm in radius. Both the top and bottom lids of the capsule measure 5.5 mm in length and radius, while the central cylinder has a length of 13.65 mm. Fig. 2(c) illustrates the air-filled leadless cardiac pacemaker device, which follows dimensions of 24.5 mm in length and 11 mm in diameter, with the upper lid having a length of 1.2 mm and the same diameter as the cylindrical body. Both wireless biomedical implants has a wall thickness of 0.25 mm. Both devices are designed using biocompatible ceramic alumina (Al_2O_3) with dielectric permittivity $\epsilon_r = 9.8$ and loss tangent ($\tan\delta = 0.006$). These devices house multiple electronic components, including batteries, sensors, electrodes, cameras, and a dual-band cubic MIMO antenna. For simulation purposes, the electronic circuitry is modeled as a substrate sandwiched between two parallel perfect electric conductor (PEC) layers. Rogers RT/droid 6010, a material with dielectric

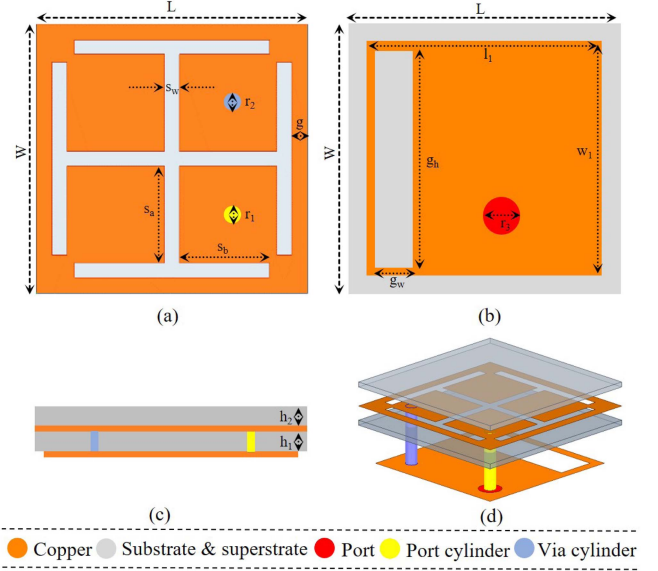


FIGURE 3. The structure of Ant-1 cubic MIMO system: (a) top view, (b) bottom view, (c) side view, (d) exploded 3D view.

permittivity $\epsilon_r = 10.2$ and a loss tangent ($\tan\delta = 0.0023$), is used for the substrate, which has a thickness of 0.4 mm.

B. SINGLE ANTENNA DESIGN AND SIMULATION SETUPS

To design a cubic MIMO system consisting of four antenna elements (Ant-I to Ant-IV) arranged on the faces of a cube, the design process commenced with the development of a single antenna element, Ant-I. Fig. 3 presents both the 2D and exploded 3D representations of Ant-1, which has overall dimensions of $4.6 \times 4.6 \times 0.26 \text{ mm}^3$. The Ant-I structure comprises multiple components, including a superstrate, substrate, shorting pin, radiating patch, and ground plane. To facilitate miniaturization, a high-dielectric constant material, Rogers RT/duroid 6010, characterized by a dielectric permittivity of $\epsilon_r = 10.2$ and a loss tangent of $\tan\delta = 0.0023$, is employed. This material is sandwiched between the primary radiating patch and a slightly slotted ground plane. Additionally, to enhance biocompatibility and mitigate the harmful potential effect on human tissue, proposed antenna is integrated within capsule devices and biocompatibility realized by employing biocompatible material. To further improve the isolation between antenna and circuitry within the capsule devices, a superstrate identical in size and material to the substrate is incorporated above the radiating patch. To minimize the overall antenna volume, the design incorporates a thin substrate with a thickness of 0.13 mm. The antenna is excited through a 50Ω coaxial feed port with a diameter of 0.6 mm. In light of the practical constraints associated with implantable devices, the Ant-I is designed to resonate at two frequencies, 1.3975 GHz and 2.45 GHz. The design of implantable antennas diverges significantly from that of antennas operating in free space, requiring specialized environments for both simulation and measurement. The encapsulated Ant-I is modeled and evaluated using a commercially available high-frequency

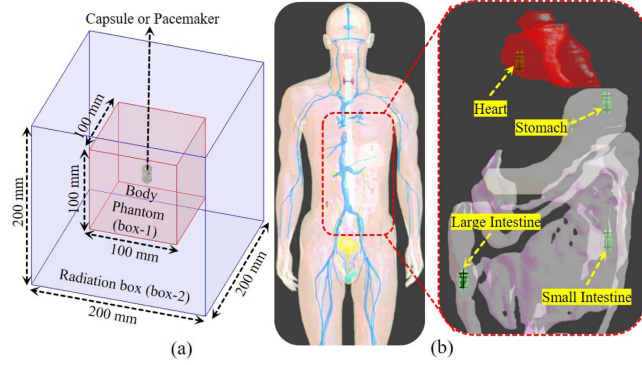


FIGURE 4. Simulation setup of the proposed cubic MIMO antenna: (a) homogeneous and (b) heterogeneous environments.

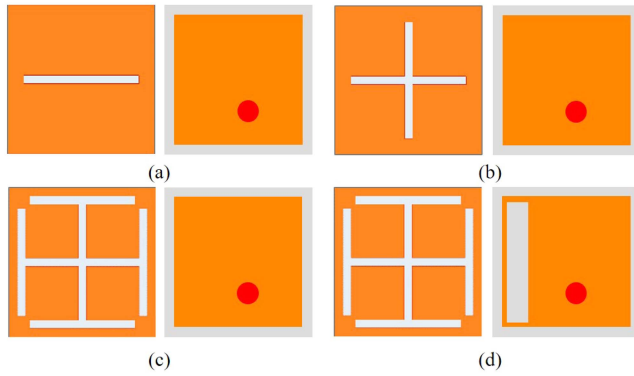


FIGURE 5. Design evolution steps of dual-band Ant-I cubic MIMO system. (a) Step I. (b) Step II. (c) Step III. (d) Step IV.

structural simulator (HFSS). In the initial simulation phase, the encapsulated Ant-I is placed at the center of a homogeneous small intestine phantom considered to be the most complicated part of the GI tract system, measuring 100 mm × 100 mm × 100 mm (referred to as box-1), and is surrounded by a radiation boundary box with dimensions of 200 mm × 200 mm × 200 mm (referred to as box-2), as illustrated in Fig. 4(a). The proposed antenna is implanted at a depth of 50 mm. In order to confirm the accuracy of the homogeneous results and get more accurate results, we simulated the antenna in heterogeneous simulation environment maintaining the same implant depth at the same position as considered in homogeneous environment. The implant depth at all position remain the same in both simulation environments to show consistency in results. Fig. 4(b) show the sim4life heterogeneous simulation environment. The dielectric properties of human tissue, which vary with frequency at 1.3975 GHz and 2.45 GHz, are applied to box-1 to more accurately mimic the characteristics of human tissue.

C. SINGLE ANTENNA DESIGN EVOLUTION AND PERFORMANCE

The Ant-I achieved the target frequency tuning and reflection coefficient $|S_{11}|$ through a sequence of four evolutionary design stages, as depicted in Fig. 5. The initial design comprised

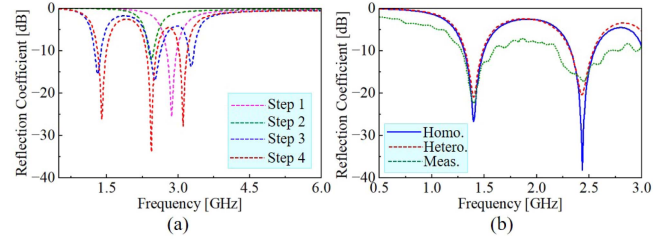
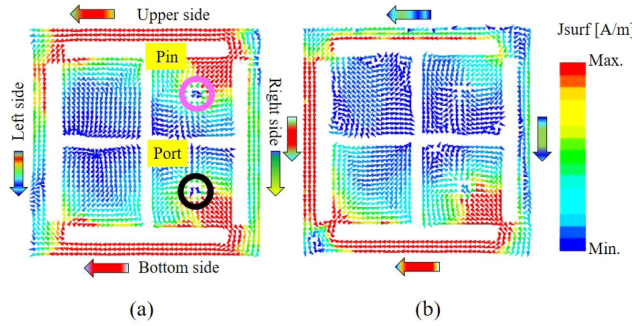


FIGURE 6. Reflection coefficient of Ant-I in homogeneous and heterogeneous small intestine phantom: (a) performance at different design evolution steps; (b) homogeneous, heterogeneous, and measured reflection coefficients of Ant-1.

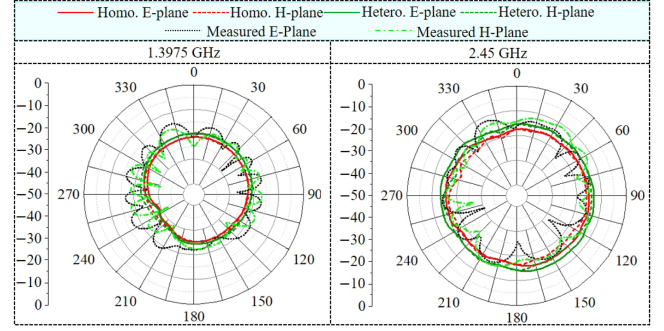
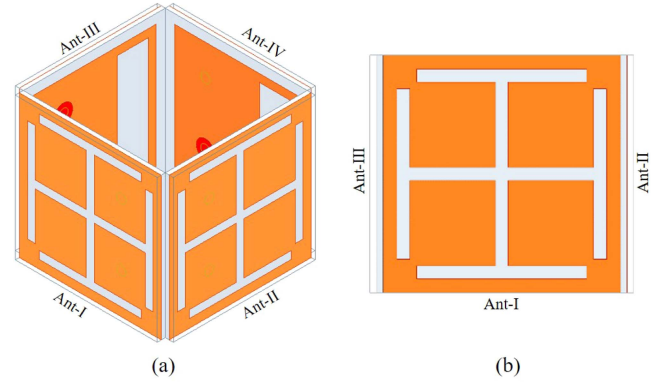
a square radiating patch featuring a horizontal rectangular slot, a complete ground plane, and a shorting pin placed opposite the feed. In the first design iteration, Ant-I exhibited resonance at 2.85 GHz, with a $|S_{11}|$ value below -25 dB as shown in Fig. 6(a). Progressing to step II, the resonant frequency shifted toward a lower range, approximately 2.43 GHz with $|S_{11}|$ below -12 dB by incorporating a vertically oriented rectangular slot of identical dimensions to the horizontal one. In step III, the design yielded a multiple-band operation, resonating around 1.31 GHz and 2.50 GHz, and 3.27 GHz with $|S_{11}|$ below -15 dB, -17 dB, and -13 dB respectively. Although the antenna resonates at three distinct frequencies in the third step, the third resonance occurs at a frequency that is significantly distant from the targeted frequency bands. The modification introduced in step III not only enabled multiband operation but also resulted in a quasi-symmetric structure of the radiating patch in all four directions, excluding the shorting pin and feed port. This near-symmetric radiating patch is advantageous in mitigating coupling effects in cubic MIMO configurations by allowing for the strategic relocation of the port and pin within the available four quadrants, along with the corresponding rotation of the ground plane. To achieve more precise antenna tuning at the desired operational frequencies, an additional rectangular slot was introduced in the ground plane, positioned opposite the port and pin. This simple adjustment effectively realized dual-band performance, with Ant-I finely tuned to 1.3975 GHz and 2.45 GHz, exhibiting $|S_{11}|$ values of -26.76 dB and -38.23 dB, respectively, as demonstrated in Fig. 6(a). Due to the compact dimensions of the designed antenna, any further modifications may compromise its dual-band performance. Additionally, such modifications could complicate the antenna structure, thereby increasing the challenges associated with its fabrication. The antenna exhibits linear polarization throughout its design evolution. As illustrated in Fig. 6(b), the homogeneous, heterogeneous, and measured results agree well with each other in terms of desired reflection coefficient and frequency tuning of a single antenna element. The optimized design parameters of the dual-band Ant-1, derived through extensive parametric studies, are detailed in Table 2. Ant-1, as part of the cubic MIMO system, exhibits peak gains of -27.5 dBi and -18 dBi at 1.3975 GHz and 2.45 GHz, respectively. To further

TABLE 2. Geometrical Parameters of Single Antenna Element

Parameters	Variables	Values [mm]
Antenna length and width	L and W	4.6
Length of vertical and horizontal slots	s_a	1.8
Width of vertical and horizontal slots	S_w	0.25
Length of edge slots	S_b	1.6
Gap between edge slot and patch	G	0.27
Substrate and superstrate height	h_1 and h_2	0.13
Length and width of ground plan	l_1 and w_1	4
Height of ground slot	g_h	3.5
Width of ground slot	g_w	1
Diameter of pin and feed cylinder	r_1 and r_2	0.3
Port diameter	r_3	0.6

**FIGURE 7. Surface current distribution of Ant-I at both resonating frequencies: (a) 1.3975 GHz, (b) 2.45 GHz.**

elucidate the behavior of the antenna, the surface current distribution across the radiating patch is plotted at both resonant frequencies. As illustrated in Fig. 7, the surface current is notably more concentrated along the edges adjacent to the feed port and pin compared to other sides of the radiating patch. At 1.3975 GHz, the current intensity is particularly high on the upper and bottom sides of the patch, while at 2.45 GHz, the current is weaker on the upper and right sides. Additionally, the current polarity at the two frequencies is the same. This high and low current concentration, coupled with the opposing polarities at two of the four edges, arises from the relative positioning of the pin and port. By relocating both, it becomes feasible to reduce coupling effects in the cubic MIMO configuration by aligning the high-current regions of one antenna with the low-current regions of another, ensuring opposite current flow directions. Further discussion, accompanied by graphical analysis, is presented in the decoupling and detuning analysis subsection. The simulated and measured radiation patterns of Ant-I at both resonant frequencies are depicted in Fig. 8. The ripples observed in Fig. 8 can be primarily attributed to the use of minced pork during experimental validation, it presents structural inhomogeneities not present in the idealized small intestine simulation environment. These irregularities lead to variations in the electromagnetic (EM) wave propagation, resulting in ripple effects in the measured results. Additionally, factors such as setup and positioning tolerances of antenna and reflections from surrounding may also contribute to these fluctuations. Despite these differences,

**FIGURE 8. Radiation pattern of Ant-I at both resonating frequencies: (left) 1.3975 GHz, (right) 2.45 GHz.****FIGURE 9. Proposed cubic MIMO configuration: (a) 3D view, (b) front view.**

the measured patterns closely follow the simulated results and confirming the design validity under realistic conditions.

D. CUBIC MIMO DESIGN

Through multiple iterative steps and conducting comprehensive investigations, the finalized Ant-I is implemented within the cubic MIMO configuration. Initially, an air-filled cube with dimensions of $4.6 \times 4.6 \times 4.6$ is created. To achieve radiation pattern diversity in four opposite directions, the Ant-I is integrated onto four opposite faces of the cube, as illustrated in Fig. 9. The cube interior remains hollow, with two faces left open to accommodate the integration of additional implantable device circuitry. The overall dimensions of the designed cubic MIMO system measure $5.12 \times 5.12 \times 4.6$ mm³, yielding a highly compact volume of 120.58 mm³. We have selected MIMO antenna size considering the dimensions of capsule like devices. Comparing antenna dimensions with capsule devices, mentioned in Section II-A, the proposed MIMO is compact enough and can easily integrate into capsule endoscopy and leadless pacemaker devices as shown in Fig. 2. Achieving precise multi-band tuning with minimum coupling effects in planar configuration presents complexity, particularly in two or more single-element-based MIMO antennas. Optimizing and tuning four antennas (Ant-I-Ant-IV) simultaneously for a cubic configuration while minimizing coupling effects and trade-offs between design parameters and

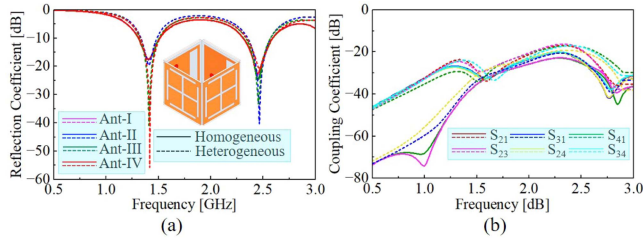


FIGURE 10. Reflection and coupling coefficients of designed cubic MIMO antenna: (a) reflection coefficient of Ant-I $|S_{11}|$, Ant-II $|S_{22}|$, Ant-III $|S_{33}|$, and Ant-IV $|S_{44}|$; (b) coupling coefficient $|S_{21}|$, $|S_{31}|$, $|S_{41}|$, $|S_{23}|$, $|S_{34}|$, and $|S_{44}|$.

performance is extremely challenging. Although the design of a single antenna element and its integration onto a cubic face offers a seemingly easy approach to developing a cubic MIMO system, the performance of the overall configuration can significantly differ from the initial antenna design. This deviation arises from significant mutual coupling between the closely spaced elements on the cube faces, an effect that is not accounted for in the design of individual antennas. To achieve the same performance in the MIMO configuration, additional structural modifications are required, which may ultimately lead to a design that differs from the original Ant-I. In this case, employing a symmetrical structural topology for the radiating patch can achieve performance equivalent to that of Ant-I without requiring modification to the antenna geometry. Fig. 10 illustrates the reflection coefficient of the cubic MIMO configuration, highlighting the mutual coupling among closely spaced adjacent elements. All four antennas (Ant-I-Ant-IV) within the cubic MIMO not only resonate at the same operational frequencies without any detuning effect but also achieve a desired reflection coefficient. The mutual coupling among the antenna elements is below -25 dB and -20 dB at 1.3975 GHz and 2.45 GHz, respectively. The proposed cubic MIMO configuration is encapsulated within a capsule device and subjected to simulation within a small intestine phantom.

E. DECOUPLING AND FREQUENCY DETUNING ANALYSIS

High coupling in MIMO antenna systems can significantly degrade overall system performance, particularly in terms of signal isolation, diversity gain, and spatial diversity. This issue is especially critical in biomedical applications, where maintaining reliable transmission quality is challenging due to the lossy nature of human body tissues. Various decoupling techniques have been proposed to mitigate coupling in MIMO systems, including stubs, strips, neutralization lines, defected ground structures, electromagnetic bandgap (EBG) structures, metamaterial structures, and spatial separation of antenna elements. However, each of these techniques has inherent limitations [28]. Recent study [25] demonstrated the use of an inductor between two antennas for biomedical applications, achieving coupling levels of -31 dB and -29 dB at two resonant frequencies. Similarly, another study [29]

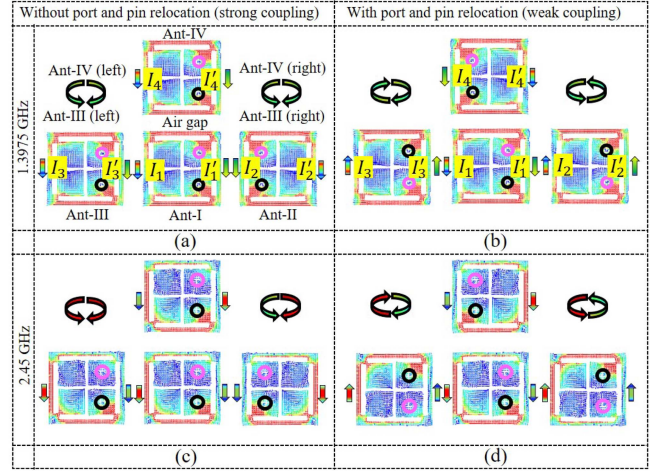


FIGURE 11. Surface current distribution of the proposed cubic MIMO antenna with and without relocation of port and shorting pin: (a),(b) 1.3975 GHz; (c),(d) 2.45 GHz.

reported isolation levels of -29.8 dB and -24.6 dB by integrating inductors between antenna elements in implantable systems. While incorporating inductors is an effective approach to reduce coupling in planar MIMO configurations, its implementation in cubic configurations, presents significant challenges. Specifically, placing inductors at the corners of the antennas in a 3D setup increases fabrication complexity and integration difficulties. These limitations highlight the need for alternative approaches in achieving effective decoupling for 3D MIMO antenna configurations. Given the space constraints in WIMBDs, a self-decoupling approach is imperative to maintain the design simplicity, miniaturization, compactness, and performance in implantable antennas. In this study, we achieved a significant reduction in coupling without introducing any modifications to the antenna structure after cubic configuration generation, without employing external components, and increasing inter-element spacing. In the proposed design, where each antenna is connected to adjacent ones on its left and right sides, while the upper and lower edges remain unconnected, as illustrated in Fig. 9. Notably, the left and right sides of Ant-I are connected to the left and right sides of Ant-II and Ant-III, respectively. At the operating frequency of 1.3975 GHz, the current distribution along the lateral sides of the antennas is relatively weak compared to the upper and lower edges, as depicted in Fig. 11(a). However, at 2.45 GHz, the current distribution shifts, with the high-current region of Ant-I connecting to the low-current sides of Ant-III and Ant-II, and vice versa, as shown in Fig. 11(c). Additionally, the current I_1 and I'_3 , I'_1 and I_2 , I_3 and I_4 , and I'_2 and I'_4 flows in the same direction across the connected sides as depicted in Fig. 11(a) and (c), generating strong currents and enhancing coupling between them at both frequencies. To address the adverse effects of high coupling, adjustments were made to the positioning of the port and pin. The symmetric design of the radiating patch offers design flexibility, allowing the port and pin to be relocated within any of the four quadrants.

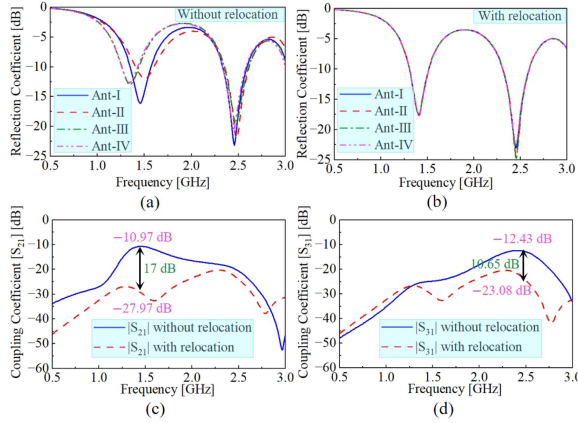


FIGURE 12. Frequency detuning and decoupling analysis of proposed cubic MIMO antenna: (a) reflection coefficient without port and shorting pin relocation, (b) with relocation, (c) $|S_{21}|$, and (d) $|S_{31}|$.

By repositioning them and appropriately rotating the ground plane, the direction of currents I_1 and I'_3 , I'_1 and I_2 , I_3 and I_4 , and I'_2 and I'_4 between the connected antennas is reversed, resulting in a substantial reduction in coupling strength at both resonant frequencies, as demonstrated in Fig. 11(b) and (d). Due to high coupling, the proposed cubic MIMO experienced a strong frequency detuning effect, especially at lower frequency, as shown in Fig. 12(a). By carefully selecting the optimal positions for the port and pin, we not only naturally mitigated the coupling effect but also maintained the same frequency tuning as achieved in the single antenna design, as illustrated in Fig. 12(b). The results clearly demonstrate that relocating them led to a substantial reduction in coupling, with $|S_{21}|$ and $|S_{31}|$ decreasing 17 dB and 10.65 dB, respectively, improving from -10.97 dB to -27.97 dB for $|S_{21}|$ and from -12.43 dB to -23.08 dB for $|S_{31}|$ at 1.3975 GHz and 2.45 GHz, as depicted in Fig. 12(c) and (d). To illustrate the impact of strategically relocating the pin and port on isolation performance, we have presented the case demonstrating the minimum isolation improvement. Notably, isolation improvements of up to 35 dB have been achieved at the corresponding resonant frequencies, as depicted in Fig. 16(b)–(g).

It is important to highlight that the proposed cubic MIMO antenna maintains excellent tuning (Ant-I–Ant-IV) and improves isolation itself without relying on external components or increased inter-element spacing. The improved isolation in this design is primarily attributed to the unique symmetric radiating patch structure and the careful positioning of the ports, pins, and ground plane. This configuration allows for the reversal of current polarities between closely connected antennas, significantly reducing mutual coupling. This technique is particularly useful for cubic MIMO configurations, where integrating external components at the corners of a cubic antenna and minimizing coupling is quite challenging. Designing and fabricating a compact, small-size antenna with a cubic configuration while ensuring good isolation

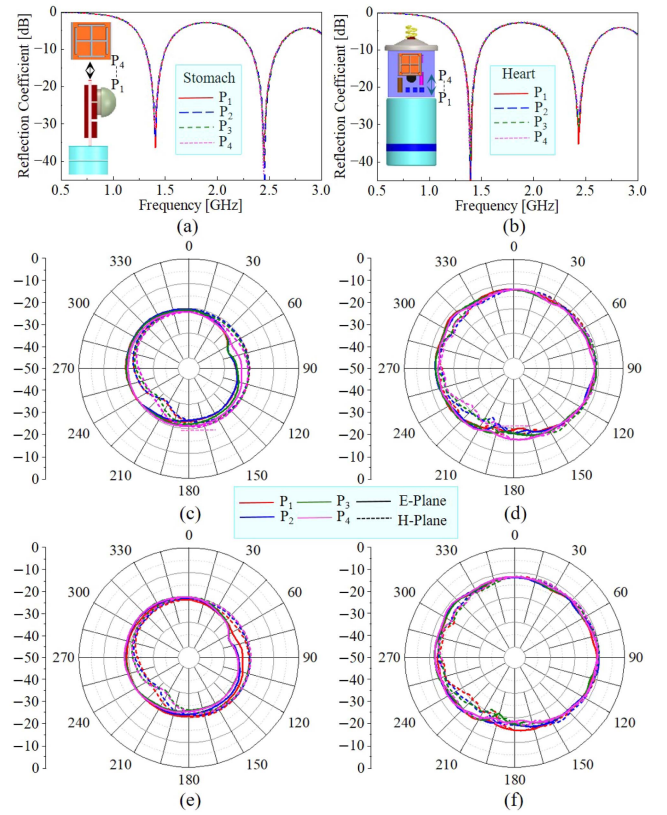


FIGURE 13. Effects of antenna location within capsule devices on antenna performance. (a) $|S_{11}|$ in capsule endoscopic. (b) $|S_{11}|$ in cardiac leadless pacemaker. (c) E and H-plane of capsule device at 1.3975 GHz. (d) E and H-plane of capsule device at 2.45 GHz. (e) E and H-plane of pacemaker device at 1.3975 GHz. (f) E and H-plane of pacemaker device at 2.45 GHz.

presents significant difficulties. Despite this, the proposed cubic MIMO antenna achieves isolation levels of -27.97 dB and -23.08 dB, well beyond the typical threshold of -15 dB. This demonstrates its effectiveness in multi-tissue applications while maintaining a compact profile.

F. CONSIDERATION OF ANTENNA LOCATION WITHIN CAPSULE DEVICES

In the context of WIMBDs, a critical consideration is the comprehensive evaluation of interaction between antenna and device components or location within the capsule device. This analysis is essential to ensure optimal performance and functionality of the device. The influence of electronic components on the S-parameters, radiation pattern, gain, and efficiency of the proposed antenna was investigated by shifting a MIMO antenna with increment 0.5 mm toward the upper lid in both devices. Fig. 13 illustrates the reflection coefficients and radiation patterns of the proposed antenna at various positions from P_1 to P_4 . For this analysis, we focused on the results of Ant-I in both capsule devices, rather than considering the results of all antenna elements. Fig. 13(a) and (b) depicts that varying the antenna position maintains resonance at both target frequencies. However, noticeable changes in the reflection coefficient were observed at both frequencies. At 1.3975 GHz,

the S_{11} values were -36.34 dB, -33.94 dB, -29.53 dB, and -30.81 dB from position P_1 to P_4 , respectively. At 2.45 GHz, the S_{11} values were -32.41 dB, -46.13 dB, -36.14 dB, and -32.37 dB, as shown in Fig. 13(a). Similar behavior was observed in the pacemaker devices, where the S_{11} ranged from -26.88 dB to -31.46 dB at 1.3975 GHz, and -22.45 dB to -23.88 dB at 2.45 GHz as shown in Fig. 13(b). In addition, Fig. 13(c)–(f) illustrate the radiation patterns of the proposed MIMO antenna at various positions from P_1 to P_4 within both capsule and pacemaker devices, evaluated at the two operating frequencies. The radiation patterns and gain remain same across different antenna locations. At 1.3975 GHz, the gain remains constant at -32.5 dBi across all positions within the capsule. At 2.45 GHz, the gain varies slightly, from -17.5 dBi to -15 dBi at positions P_2 and P_4 . A similar behavior is observed in the pacemaker device, where the gain at 1.3975 GHz remains at -32.5 dBi across all positions, while at 2.45 GHz, it slightly increases from -17.5 dBi to -15 dBi at position P_1 . In addition, we investigated the radiation efficiency, which is a critical performance parameter for implantable antennas operating in the highly lossy biological environment. Within the capsule device, the efficiency at 1.3975 GHz is -37.7 dB, -37.9 dB, -38.0 dB, and -38.4 dB for positions P_1 to P_4 , respectively. At 2.45 GHz, the efficiencies are -23.6 dB, -23.8 dB, -23.7 dB, and -24.2 dB. Similar behavior is observed in the pacemaker device, where the efficiency at 1.3975 GHz ranges from -36.8 dB to -37.5 dB, and from -22.7 dB to -23.3 dB at 2.45 GHz. Through this analysis, we confirmed that proposed antenna can effectively resonate at targeted frequencies with good $|S_{11}|$ value at any location within in the capsule devices as well as maintain other performance parameters. For optimal antenna performance, the proposed MIMO antenna was positioned adjacent to the capsule circuitry, with a sufficient gap between them.

G. EQUIVALENT CIRCUIT ANALYSIS

In order to provide a thorough analysis of the antenna performance, we have designed equivalent circuit model using Keysight Advanced Design System (ADS), as shown in Fig. 14. The proposed model represents each individual element of the MIMO system through impedance-matching and parallel RLC circuits, with a $50\ \Omega$ terminal used for excitation. The SISO circuit, which includes components C_1 , C_2 , L_1 , and L_2 , is specifically designed to achieve impedance matching across the operational bandwidth. To further optimize resonance and the reflection coefficient, an additional RLC circuit comprising L_3 , C_3 , R_1 , and L_4 is presented. The mutual coupling between the MIMO elements is modeled using a series combination of a capacitor C_4 and an inductor L_5 , which is influenced by the gap between the antenna elements. It is important to note that the antenna elements exhibit identical structural configurations and performance characteristics, and therefore, the equivalent circuit model is consistent across all elements. As depicted in Fig. 10, the proposed antenna elements resonate at the target frequencies without detuning. To

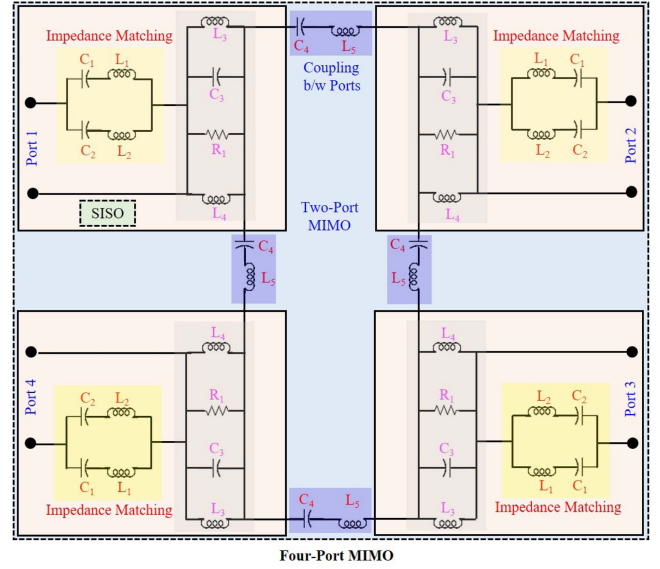


FIGURE 14. Equivalent circuit models of the SISO and MIMO ($C_1 = 0.91$ pF, $C_2 = C_3 = 0.5$ pF, $L_1 = 14.4$ nH, $L_2 = 8.42$ nH, $L_3 = 2.2$ nH, $R_1 = 3.5\ \Omega$, $L_4 = 0.37$ nH, $C_4 = 0.63$ pF, $L_5 = 2$ nH).

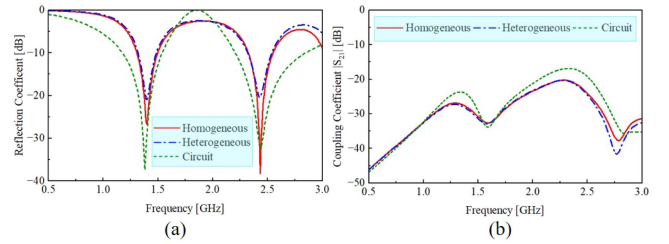


FIGURE 15. Scattering parameters of designed cubic MIMO antenna: (a) reflection coefficient, (b) coupling coefficient.

enhance clarity, the individual antenna element results are presented in Fig. 15. These results, which included simulations in various environments such as homogeneous, heterogeneous, and circuit model, demonstrate consistency, showing good agreement with each other and overall performance in terms of coupling, resonance, and reflection coefficient.

III. SIMULATION, FABRICATION, AND MEASURED RESULTS

A. SCATTERING PARAMETERS IN HEART AND VARIOUS GI TRACT LOCATIONS

The designed cubic MIMO antenna demonstrated stable frequency tuning and minimal coupling effects, enabling its placement in various body tissues, including the entire GI tract region and the heart, to support applications such as capsule endoscopy and cardiac leadless pacemaker systems. Fig. 16 presents the reflection and coupling coefficients of the proposed cubic MIMO antenna across different conditions, including its placement in the homogeneous and heterogeneous small intestine, large intestine, stomach, and heart. The reflection coefficients vary between -17 dB to -40 dB at 1.3975 GHz and -18 dB to -32 dB at 2.45 GHz. Despite

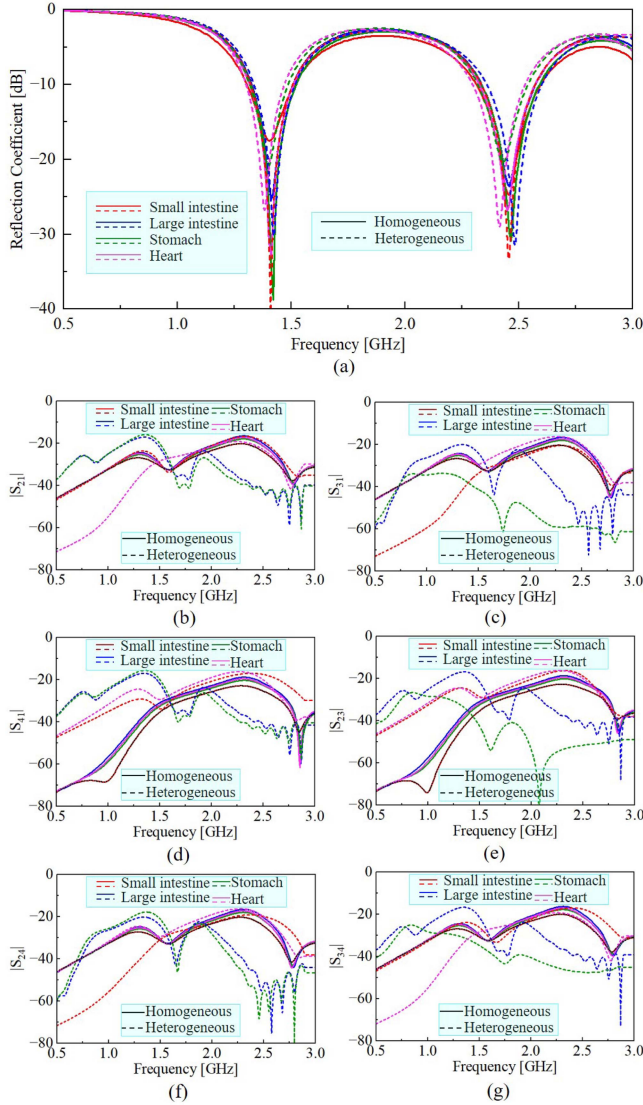


FIGURE 16. Scattering parameters of cubic MIMO antenna in different human body tissues: (a) reflection coefficient, (b)–(g) coupling coefficient.

this, proposed MIMO, resonates effectively within the target frequency bands, maintaining desired reflection coefficients across all simulated environments as shown in Fig. 16(a). Similarly, the coupling coefficients $|S_{21}|$, $|S_{31}|$, $|S_{41}|$, $|S_{23}|$, $|S_{24}|$, and $|S_{34}|$ vary between -20 dB to -40 dB at 1.3975, while at 2.45 GHz, it ranges from -20 dB to -70 . Due to symmetry of coupling coefficients $|S_{12}| = |S_{21}|$, $|S_{13}| = |S_{31}|$, $|S_{14}| = |S_{41}|$, $|S_{23}| = |S_{32}|$, $|S_{24}| = |S_{42}|$, and $|S_{34}| = |S_{43}|$, we have considered only $|S_{21}|$, $|S_{31}|$, $|S_{41}|$, $|S_{23}|$, $|S_{24}|$, and $|S_{34}|$. All coupling coefficients are remain below -20 dB across all phantoms at both resonant frequencies, as illustrated in Fig. 16(b)–(g). Furthermore, coupling levels as low as -40 dB, -50 dB, and even -60 dB were achieved. Antenna specifications inherently vary under different implantation conditions due to changes in the surrounding environment. Nevertheless, despite these variations, the proposed MIMO antenna remain resonates at the intended frequency bands and maintains good performance in terms of reflection and

TABLE 3. Human Body Tissue Properties at Resonant Frequencies

Frequency	1.3975 GHz		2.45 GHz	
Tissue	Permittivity (ϵ_r)	Elec. Cond. (S/m)	Permittivity (ϵ_r)	Elec. Cond. (S/m)
Heart	57.5	1.51	54.8	2.26
Stomach	63.9	1.44	62.2	2.21
Small Intestine	57.1	2.44	54.4	3.17
Large Intestine	56.1	1.33	53.9	2.03

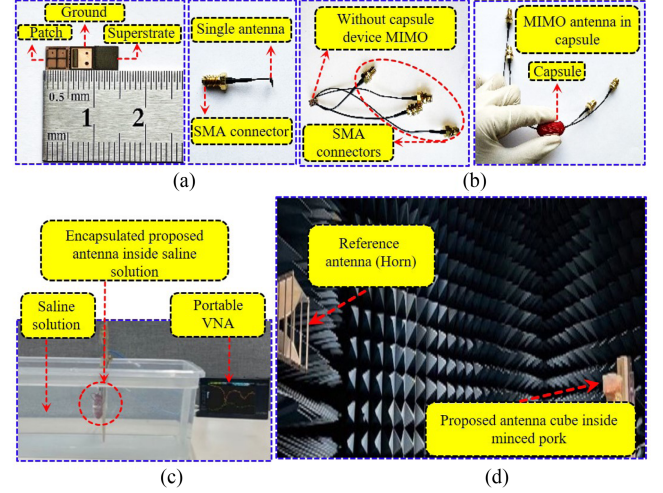


FIGURE 17. (a) Fabricated prototype of SISO, (b) prototype of cubic MIMO antenna with and without encapsulation device, (c) and (d) experimental setup for scattering parameters and radiation pattern respectively.

coupling coefficients. The dielectric properties, including permittivity and electrical conductivity of the tissues at both operational frequencies, are summarized in Table 3.

B. FABRICATION AND MEASURED SCATTERING PARAMETERS

To validate the simulation results, the proposed antenna is fabricated as shown in Fig. 17(a). The prototypes of the proposed MIMO antenna are encapsulated within a 3D-printed capsule-like structure as illustrated in Fig. 17(b). The fabricated prototypes of the proposed SISO and MIMO antennas are tested within saline solution as shown in Fig. 17(c). The saline solution was carefully prepared by adjusting the concentrations of salt and sugar in water to replicate the characteristics of tissue. This method is commonly employed in experimental studies involving implantable antennas. Several recent studies has considered saline solution for scattering parameters measurements [20], [24]. In order to get a measured results close to the simulation ones, the same implant depth has been maintained as considered in both simulation environments. The saline box used in the experimental setup has dimensions of 300 mm $\times$ 160 mm $\times$ 100 mm. The capsule device is positioned near the center of the box, fully immersed in the saline solution. The proposed antenna is located at the bottom of the capsule, corresponding to half the height of the box (50 mm implant depth). The scattering parameters, including reflection and coupling coefficients, are measured using a two-port vector network analyzer (VNA) connected via a 50 Ω

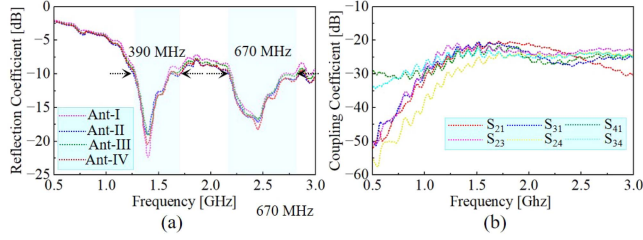


FIGURE 18. Measured performance of proposed cubic MIMO antenna: (a) reflection coefficient, (b) coupling coefficient.

flexible subminiature-A (SMA) cable. The proposed MIMO antenna is tested by connecting two of its ports to the VNA while the remaining ports are terminated with matched loads (50Ω), and the process is repeated for all ports. As illustrated in Fig. 18, all four antennas of the fabricated cubic MIMO exhibited reflection coefficients $|S_{11}|$, $|S_{22}|$, $|S_{33}|$, and $|S_{44}|$ below -22 dB, -18 dB, -19 dB, and -20 dB at 1.3975 GHz and -18 dB, -18 dB, -17 dB, and -19 dB at 2.45 GHz. Minor discrepancies between the ideal design in the simulated environment and the fabricated prototype in the real-time environment resulted in slightly less smooth measured S-parameters curves compared to the ideal simulation results. Nonetheless, the measured results achieved the same resonant frequencies and the desired reflection coefficient levels with coupling below -23 dB and -24 dB, respectively. Additionally, the fabricated MIMO antenna demonstrated wide bandwidths of approximately 390 MHz and 670 MHz, with fractional bandwidth percentages of more than 27% and 26% at the lower and higher resonance frequencies, respectively. For measured results an experimental setup was carefully employed. The setup was positioned on a non-reflective platform, and measurements were conducted in such environment to reduce undesired reflections. Despite minor ripple effects the measured performance agree with simulation results validating the effectiveness of the antenna design under realistic and practical conditions.

C. SIMULATED AND MEASURED RADIATION PATTERNS

The far-field radiation patterns of the proposed antenna are experimentally evaluated within an anechoic chamber. A high-gain reference antenna (horn antenna) was positioned at a distance of 10 m from the proposed MIMO antenna inside minced pork. For gain and radiation pattern measurements, minced pork was employed instead of saline solution. This was chosen to avoid potential leakage and maintain setup stability during system rotation along the θ - and ϕ -axis. The minced pork offered a more stable medium for far-field measurement, enabling consistent alignment with the reference antenna. During the measurements, the one port of the presented MIMO is connected to a VNA while the reference antenna is connected to the rest of the VNA ports. The remaining ports of the MIMO antenna are terminated with matched loads (50Ω), and the procedure is repeated for all antenna elements. The simulated and measured radiation patterns for each

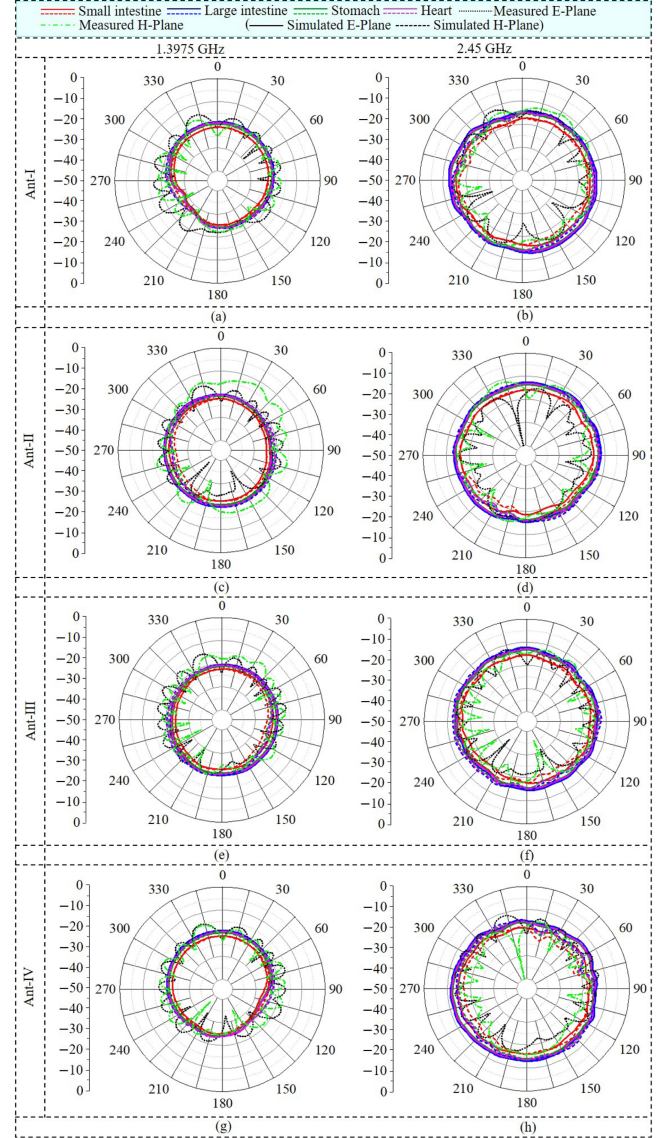


FIGURE 19. Normalized radiation pattern of all antenna elements in cubic MIMO configuration at both resonant frequencies: (a),(b) Ant-I, (c),(d) Ant-II, (e),(f) Ant-III, and (g),(h) Ant-IV.

element of the cubic MIMO configuration at both resonant frequencies are presented in Fig. 19, and both are showing good agreement with each other. At 1.3975 GHz, Ant-I primarily radiates between the x - and y -planes, whereas at 2.45 GHz, it exhibits a quasi-omnidirectional radiation pattern, as shown in Fig. 19(a) and (b). In contrast, Ant-IV radiates in directions opposite to Ant-I due to the reversed orientation of its port and pin, as illustrated in Fig. 19(g). Meanwhile, Ant-II and Ant-III radiate in the remaining directions but oppositely to each other. The cubic MIMO configuration, thus, facilitates equal power radiation in all directions, which is quite unattainable in planar configurations. The simulated peak gains of the antenna within biological environments are found to be -32.5 dBi, -30 dBi, -32.5 dBi, and -37.5 dBi at 1.3975 GHz for the heart, stomach, large intestine, and small intestine, respectively. Similarly, at 2.45 GHz, the simulated peak gains were

−17.5 dBi, −20 dBi, −17.5 dBi, and −22.5 dBi for these respective regions. The reduced gain in the small intestine is attributed to the significant attenuation of EM waves caused by surrounding biological tissues, which are highly lossy and possess high water content. Furthermore, the small intestine complex structure and variable internal contents introduce additional challenges to signal propagation, further contributing to attenuation. The measured gain of the proposed MIMO antenna is −28.17 dBi and −18.15 dBi at lower and higher resonant frequencies, respectively. At lower frequency 1.3975 GHz, the antenna gains are comparatively lower than 2.45 GHz. Biological tissues like small intestine, large intestine, stomach, and heart exhibit higher dielectric losses due to higher dielectric permittivity values at lower frequency as shown in Table 3, which leads to greater energy loss within the body rather than efficient radiation. As a consequence, the signal is absorbed more by the surrounding tissues, thereby reducing the antenna gain. Antenna efficiency is a critical factor in ensuring effective transmission, especially in the context of implantable devices. When an antenna is placed within the human body, which is a highly lossy medium due to the presence of biological tissues, its efficiency often decreases significantly. This is primarily due to absorption and strong coupling with surrounding tissues. Additionally, the miniaturization required for implantable antennas tends to increase losses and reduce the effective aperture of the antenna, further compromising its efficiency. Typically, in-body implantable antennas exhibit efficiencies of approximately 1% or less, depending on the specific type of tissue and implant depth [31]. For the proposed MIMO antenna, the radiation efficiencies are −30.96 dB and −23.01 dB, at both lower and higher resonant frequencies respectively, when evaluated at an implant depth of 50 mm. These values were calculated using a small intestine phantom, as this region is known to produce significant attenuation of EM waves due to the high water content and lossy nature of the surrounding tissues. Fig. 19, the antenna elements have quite distinct radiation patterns, a characteristic that plays an important role in lowering the ECC within both frequency bands. This reduction in ECC contributes to an improvement in the system overall channel capacity. Additionally, introduce the capability to be coupled with rectifier circuits, enabling them to collect energy from various directions. This feature addresses the challenge of alignment between the wireless power transmitter and the receiving antenna, ensuring more stable power transmission. Consequently, the system power transfer efficiency is significantly enhanced. Thus, the proposed configuration offers two primary advantages: first, as a transmitter, it improves channel capacity by leveraging the diversity of radiation, and second, as a receiver, it enhances wireless power transfer efficiency by absorbing energy from multiple directions.

D. EFFECT OF IMPLANT DEPTH ON ANTENNA GAIN

Implant depth significantly influences antenna performance, particularly its gain. Increasing the implant depth results in a decrease in antenna gain. As the implant depth increases,

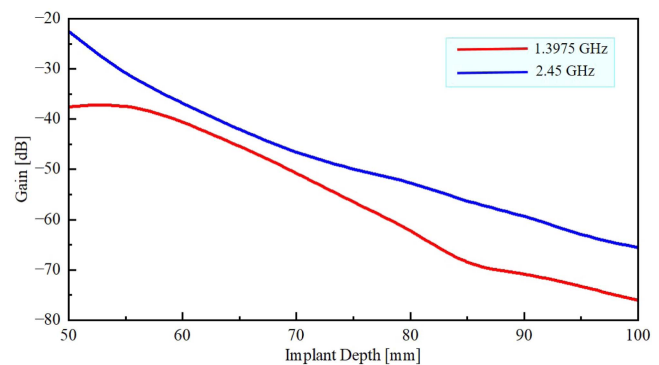


FIGURE 20. Antenna gain analysis at various implant depth of the proposed MIMO antenna in small intestine phantom at both resonant frequencies.

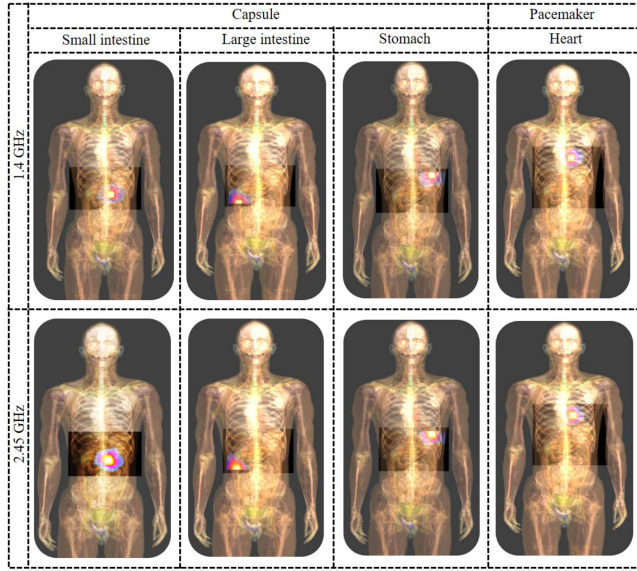
the signal must propagate through a large volume of tissue, leading to higher signal attenuation. This is because the wave propagation encounters more material with each additional millimeter of depth, which weakens the signal. Tissues with higher water content, such as those in the stomach and intestine, tend to absorb and scatter signals more effectively, further contributing to signal loss. To evaluate the impact of implant depth on antenna gain, we varied the depth from 50 mm to 100 mm. Fig. 20 shows that as the implant depth increases, antenna gain decreases at both resonant frequencies. At a depth of 100 mm, the gain drops to a minimum of −76.02 dBi at the lower frequency and −65.53 dBi at the higher frequency. Similarly, at 50 mm, the gain is −37.5 dBi and −22.5 dBi, respectively. The low gain at higher implant depths is attributed to the high conductivity of the small intestine, which increases attenuation. Despite this, the antenna operates effectively at the considered implant depths. To demonstrate the impact of implant depth on gain, we focused on the case of small intestine tissue, which exhibits the lowest gain among other body tissues due to its high conductivity. Overall, the proposed antenna performs well across the tested depth range, making it suitable for deep tissue applications in various body regions investigated in this work.

E. SAR EVALUATION FOR SAFETY INSURANCE

In order to ensure user health safety, it is important to evaluate the potential risks associated with the proposed implantable MIMO antenna. The SAR is a fundamental measure for evaluating the safety consequences of EM wave exposure on human tissue. Following IEEE standard IEEE (C95.1-2019), the permissible maximum averaged SAR for tissue samples weighing 1-g and 10-g should not exceed 1.6 W/Kg and 2 W/Kg, respectively [30]. For the calculation of SAR values, a variety of commercially available simulation tools have been employed in recent years. Notable examples include the heterogeneous human voxel model (Gustav) in CST, the 3D human voxel model in Remcom, and Sim4Life. Several recent studies employed Sim4Life for SAR evaluation due to its ability to provide highly realistic human body simulations

TABLE 4. SAR (W/kg) and the Corresponding Maximum Allowable Input Power (MAIP) (mW) for the Proposed Cubic MIMO Antenna System

Maximum SAR (W/kg) and MAIP for 10-g (at 1 watt input power)				
Frequency	1.4 GHz		2.45 GHz	
	SAR (W/kg)	Input power (mW)	SAR (W/kg)	Input power (mW)
Small intestine	90.4	22.12	87.8	22.7
Large intestine	90.5	22.0	87.3	22.9
Stomach	90.1	22.1	87.2	22.9
Heart	90.5	22.0	89.4	22.3

**FIGURE 21.** Normalized 10-g SAR distribution of the proposed MIMO antennas in different tissues at 1.4 GHz and 2.5 GHz.

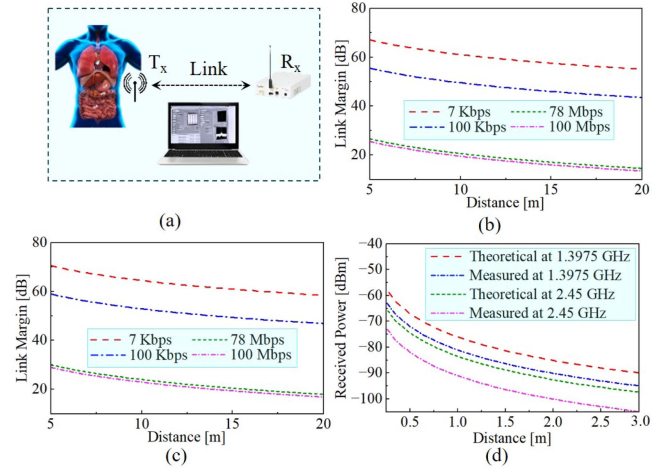
environments [25]. For the evaluation of the SAR, we employed Sim4Life software, owing to its advanced and realistic simulation of human body environments. Within the scope of this study, the implantable MIMO antenna is placed within the heterogeneous structures of the heart, stomach, large intestine, and small intestine of realistic human phantoms as shown in Fig. 4(b). Each antenna element is stimulated with an input power of 1 W [24], [25], [26]. The SAR values, along with the maximum allowable power for the proposed implantable antenna at the two resonant frequencies, are detailed in Table 4. Furthermore, Fig. 21 depicts the normalized SAR distributions across in all phantoms, maintaining the same depth as the homogeneous phantoms for both frequency bands. Given the restricted input power constraints for biomedical implants (−16 dBm), the proposed antenna demonstrates safety across both operational frequencies.

F. WIRELESS TELEMETRY COMMUNICATION ANALYSIS

The wireless telemetry function of the proposed antenna is rigorously assessed by conducting an evaluation of its transmission performance, employing a combination of theoretical and experimental frameworks. Table 5 lists the parameters used for the evaluation of the communication links. In up-link

TABLE 5. Key Parameters for the Analysis of Wireless Communication Link of Proposed Antenna

Parameters	Variables	Unit	Values
Frequency	F	GHz	1.3975, 2.45
Transmitter power	P_t	dBm	−16
Antenna gain	G_t	dBi	−17
Receiver gain	G_r	dBi	2
Distance	D	m	0-20
QPSK	E_b/N_0	dB	9.6
Boltzmann constant	k	---	1.38×10^{-23}
Temperature	K	T_0	273
Noise power density	N_0	dB/Hz	203.93

**FIGURE 22.** (a) Conceptual model of wireless communication link analysis, (b) link margin at 1.3975 GHz, (c) link margin at 2.45 GHz, (d) theoretical and experimental received power.

wireless communication, the proposed antenna functions as the transmitting T_x element, whereas an external antenna is designated as the receiving R_x component. The theoretical derivation of the wireless link performance is governed by the following equations:

$$LM = P_a - P_r \quad (1)$$

where LM represent link margin, P_a is available power, and P_r is a received power. The available power can be calculated as follow:

$$P_a = P_T + G_T + G_R - L_p - PL \quad (2)$$

where PL , is the path loss, can be expressed as follows:

$$PL(dB) = 10\gamma \log_{10} \left(\frac{d}{d_0} \right) + 20 \log_{10} \left(\frac{4\pi d_0}{\lambda_0} \right) + X_\sigma \quad (3)$$

In this expression, γ represents the path loss exponent, d_0 denotes the reference distance where $d_0 < d$, d corresponds to the separation between T_x and R_x , λ_0 signifies the free-space wavelength, and X_σ accounts for the shadowing factor. The conceptual model of the wireless communication link is shown in Fig. 22(a). As illustrated in Fig. 22(b) and (c), for data rates of 7 Kbps, 100 Kbps, 78 Mbps, and 100 Mbps at 1.3975 GHz and 2.45 GHz, the proposed MIMO antenna can establish high speed (100 Mbps) reliable communication

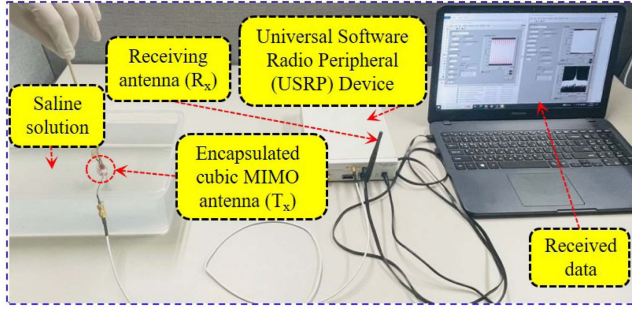


FIGURE 23. Experimental setup for measured received power.

link in a radius of 10 m and 15 m at both frequencies, respectively. Additionally, to compute the received power in link budget analysis, we conducted experimental analysis as shown in Fig. 23. The configuration involved a container of saline solution, the proposed antenna acting as a T_x , an external monopole antenna as a R_x , a Universal Software Radio Peripheral (USRP) device, and a laptop interfaced with LabVIEW software to monitor the received power data. It is important to note that the received power is determined by connecting the USRP T_x port to a single antenna element. The experiment commenced with the monopole antenna functioning as the R_x , positioned at a distance of 0.3 m from the saline-filled container. The modulation scheme employed for the experiment is quadrature phase shift keying (QPSK), with a sampling rate of 500K samples per second. Data transmission is packet-based, where each symbol is composed of eight samples. Given the USRP receiver sensitivity of -112 dBm, the experiment is conducted on indoor propagation, with the distance between the T_x and R_x antennas varying incrementally from 0.3 m to 3 m. A comparative analysis of the theoretical and experimentally measured received power levels across the relevant frequencies is presented in Fig. 22(d). Both the simulation and experimental results confirm that the proposed implantable antenna system can maintain communication over distances up to 3 m in both frequency bands, with a received power exceeding -100 dBm. It is important to note that the received power measurements were conducted using a saline solution. In future studies, we intend to perform a more rigorous investigation into the influence of realistic phantoms compared to saline solution on the received power performance of the antenna system.

G. RADIATION DIVERSITY CHARACTERISTICS

Fig. 24 illustrates the radiation diversity and the maximum and minimum gain with calculated deviations of 18.12 dB and 20.96 dB at both lower and higher resonant frequencies respectively. These gain values correspond to the small intestine, a tissue that exhibits higher loss compared to other tissues, leading to the observed deviations in gain. Despite these variations, the four antenna elements, each radiating in distinct and opposing directions, ensure radiation diversity

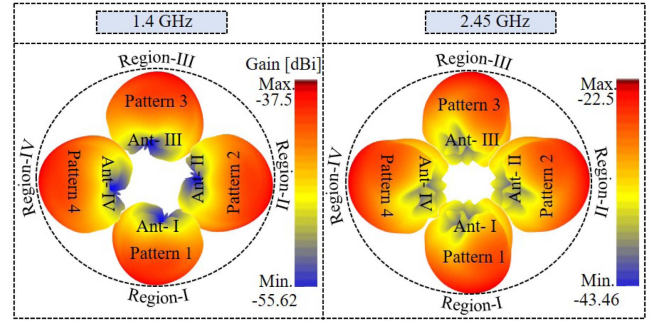


FIGURE 24. Maximum radiation direction of proposed cubic MIMO antenna in respective regions.

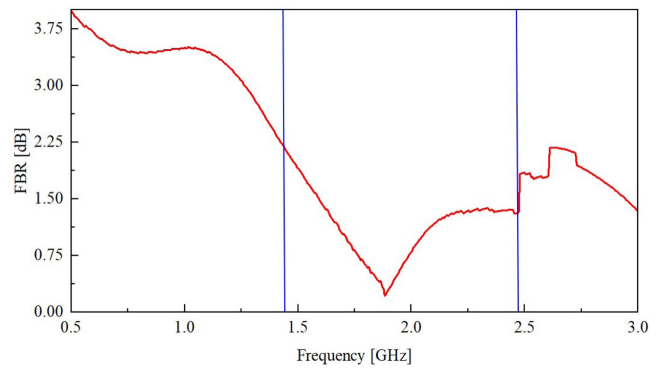


FIGURE 25. Front to back ratio of individual antenna element.

across all four directions. Additionally, the cubic configuration of the antenna provides isotropic coverage, offering a significant advantage over traditional 2D planar MIMO systems, which are inherently directional and typically limited to maintaining communication links in a single direction. In contrast, the cubic MIMO enables omnidirectional communication with greater range, ensuring good and highly stable communication links across all directions. This capability effectively addresses challenges associated with continuous movement and varying orientations of implantable devices, particularly capsule endoscopy, where the encapsulated antenna moves through the complex and dynamic GI tract system. The comprehensive radiation pattern and isotropic coverage significantly enhance the reliability, efficiency, and adaptability of wireless biomedical systems in complex in-body applications. Although the radiation patterns originating from Ant-I, Ant-II, Ant-III, and Ant-IV appear to be directive in Fig. 24, this is not the case. To further clarify, we calculated the Front-to-Back Ratio (FBR) of Ant-I at both operating frequencies. The FBR values at these frequencies are 2.40 dB, and 1.30 dB as shown in Fig. 25. Since the FBR values of the proposed antenna at both frequencies are well below 2.25 dB, this confirms that the antenna exhibits omnidirectional-like behavior as shown in Fig. 19.

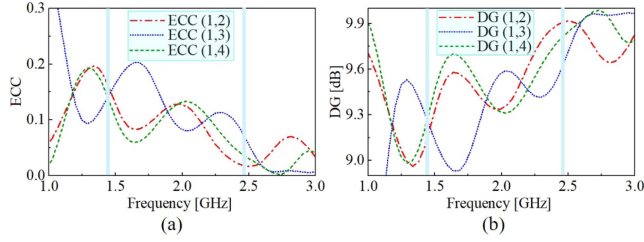


FIGURE 26. Channel parameters of the proposed cubic MIMO antenna at both resonant frequencies: (a) ECC and (b) DG.

IV. MIMO CHANNEL PARAMETERS

The evaluation of a MIMO antenna system, characterized by multiple antenna elements, extends beyond the assessment of antenna isolation typically required for single antenna designs. A critical factor in determining its viability lies in the evaluation of the inter-element correlation within the antenna system. Fundamental performances, such as the ECC, DG, CCL, and TARC, play a significant role in establishing the degree of independence among the antenna elements in MIMO configurations. These parameters are indispensable for thoroughly validating the system performance and efficiency.

A. ENVELOP CORRELATION COEFFICIENT (ECC)

The ECC calculates the isolation or correlation between antenna elements in close proximity within a MIMO system. Ideally, the antenna elements should exhibit zero correlation; however, in practical scenarios, achieving such ideal conditions is unattainable. As a result, an ECC value below 0.5 is generally deemed acceptable. It can be derived using either S-parameters or radiation patterns. While S-parameters are frequently employed to calculate ECC, their applicability is limited to lossless antennas operating in isotropic environments. This assumption is often overlooked, leading to inaccurate ECC estimates, particularly because printed antennas tend to be lossy, especially in complex environments like the human body. Conversely, the radiation pattern method offers a more precise representation, as it directly relates to the actual channel behavior between the transmitter and receiver, accounts for lossy antennas, and accurately calculates spatial characteristics and mutual coupling effects. Additionally, the radiation pattern approach considers the correct radiation properties and their influence on channel performance, making it a more reliable method for practical evaluation. Consequently, in this study, the ECC is computed from the far-field radiation patterns of the proposed antenna using the following equation:

$$\text{ECC} = \frac{\left| \int_0^{4\pi} \vec{A}_{xi}(\theta, \phi) \times \vec{A}_{yj}(\theta, \phi) d\Omega \right|^2}{\int_0^{4\pi} |\vec{A}_{xi}(\theta, \phi)|^2 d\Omega \int_0^{4\pi} |\vec{A}_{yj}(\theta, \phi)|^2 d\Omega} \quad (4)$$

where $\vec{A}_{xi}$ and $\vec{A}_{yj}$ denote the three-dimensional radiation patterns of the i th and j th antenna elements, while Ω represents the solid angle. Fig. 26(a) illustrates the computed ECC values across the respective frequency bands. At lower

resonant frequency, the ECC values of the proposed antenna among ports 1, 2, 3, and 4 are 0.19, 0.11, and 0.17, while at higher resonant frequency they are 0.02, 0.08, and 0.04, as shown in Fig. 26(a). Throughout these two bands, the ECC consistently remains below 0.2 and 0.1, respectively. These remarkably low ECC values highlight the excellent isolation performance of the proposed MIMO antenna system, making it an ideal candidate for IMDs that demand high data-rate communication.

B. DIVERSITY GAIN (DG)

The DG is another significant parameter evaluating the improvement signal-to-interference ratio that arises from the implementation of diversity techniques, thereby ensuring the antenna performance without losses. An optimal DG value of 10 dB is typically regarded as indicative of zero correlation among the antenna elements. The calculation of DG can be performed using the following equation:

$$\text{DG} = 10\sqrt{1 - \text{ECC}^2} \quad (5)$$

The DG exhibited by the proposed MIMO antenna is illustrated in Fig. 26(b). At lower resonant frequency, the DG values among ports 1, 2, 3, and 4 are 9.2 dB, 9.3 dB, and 9.25 dB, respectively. At higher frequency, the DG value is 9.8 dB, 9.5 dB, and 9.7 dB among antenna ports, as shown in Fig. 26(b). The value of a DG exceeding 9.2 dB within the targeted frequency bands demonstrates the excellent performance. This performance ensures the antenna suitability for scenarios requiring robust communication capabilities.

C. CHANNEL CAPACITY LOSS (CCL)

The CCL represents a critical parameter in MIMO systems, defining the ultimate limit on the data transmission rate achievable across the communication channel. It is calculated as follows:

$$\text{CCL} = -\log_2 \det(\alpha^R) \quad (6)$$

where α^R denotes the correlation matrix, defined as follows [18]:

$$\alpha^R = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ R_{31} & R_{32} & R_{33} & R_{34} \\ R_{41} & R_{42} & R_{43} & R_{44} \end{bmatrix} \quad (7)$$

The diagonal R_{ii} and off-diagonal R_{ij} elements of the correlation matrix α^R are provided as follows:

$$R_{ii} = 1 - \sum_{j=1}^N |S_{ij}|^2 \text{ and } R_{ij} = -(S_{ii}^* S_{ij} + S_{ji}^* S_{jj}) \quad (8)$$

for $i, j = 1, 2, 3, 4$.

The CCL of the proposed MIMO implantable antenna is illustrated in Fig. 27(a). The CCL values are measured at 0.04 b/sec/Hz and 0.12 b/sec/Hz at the lower and higher resonant

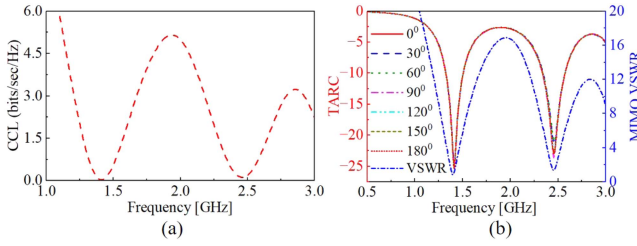


FIGURE 27. Channel parameters of the proposed cubic MIMO antenna: (a) CCL and (b) TARC and VSWR.

frequencies, respectively. The figure indicates that the CCL remains below the set threshold of 0.4 b/sec/Hz [18], signifying improved performance of the MIMO system.

D. TOTAL ACTIVE REFLECTION COEFFICIENT (TARC)

The TARC is another essential parameter in MIMO systems, used to evaluate the proportion of power that is reflected back relative to the incident power at each antenna element. TARC is defined as the square root of the ratio of total reflected power to total incident power across all antenna elements. This parameter is significant for evaluating MIMO antenna performance during simultaneous multi-port operations, determining the operational frequency bands of the MIMO system, and confirming radiation performance under varying input phases. For the proposed four-port cubic MIMO configuration, TARC can be calculated as follows:

$$\text{TARC} = \frac{\sqrt{\sum_{n=1}^N |S_{i1} + \sum_{m=2}^N S_{im}^{e^{j\theta_{m-1}}}|^2}}{\sqrt{N}} \quad (9)$$

The TARC values corresponding to various excitation phases (0° , 30° , 60° , 90° , 120° , 150° , and 180°) are presented in Fig. 27(b). The TARC remains below -10 dB at both resonant frequencies, which indicates stable MIMO performance. Furthermore, the voltage standing wave ratio (VSWR) for the proposed cubic MIMO is maintained below 2, further validating its suitability for dual-band functionality, as depicted in Fig. 27(b). Based on the evaluation of the MIMO channel parameters, the proposed antenna exhibits substantial MIMO performance, establishing it as a promising candidate for various high-data-rate biomedical applications, including capsule endoscopy and leadless pacemaker systems with radiation diversity.

V. CONCLUSION

A highly efficient, compact, dual-band, cubic MIMO antenna with high data rates has been proposed for wireless capsule endoscopy and cardiac leadless pacemaker implants. The proposed antenna covers two frequency bands, including 1.395 GHz to 1.4 GHz and 2.4 GHz to 2.4835 GHz. Antenna miniaturization is achieved with the inclusion of symmetric radiating patch slots as well as coupling reduced by strategically positioning the pin, port, and ground slot.

The proposed cubic MIMO antenna with four individual elements has overall dimensions of $5.12 \times 5.12 \times 4.6 \text{ mm}^3$, yielding a highly compact volume of 120.58 mm^3 . To ensure realistic evaluation and assess safety for human implantation, the encapsulated antennas was simulated in heterogeneous environments, and the SAR values were calculated by locating it within different realistic human tissues, including the small intestine, large intestine, stomach, and heart. Results confirm that the proposed MIMO antenna is extremely safe for practical biomedical applications. A fabricated prototype tested in minced pork and measured results demonstrated good performance, including reflection coefficient (-22 dB, -19 dB), gain (-28.17 dBi, -18.15 dBi), -10 dB bandwidth (390 MHz, 670 MHz), minimal coupling coefficient (-23 dB, -24 dB), and fractional bandwidth (27%, 26%) at 1.3975 and 2.45 GHz, respectively, as well as agreeing well with simulation results. Furthermore, the antenna established a high-speed 100 Mbps reliable communication link, enabling efficient high-quality data transmission in a radius of up to 10 m and 15 m at the lower and higher frequencies, respectively. In addition, the proposed cubic MIMO achieved a radiation diversity in all direction, equally covering four regions. The key MIMO channel parameters, including ECC, DG, CCL, and TARQ, were analyzed, further validating the effectiveness of the presented MIMO design. The proposed radiation diversity-enabled, self-isolated, compact, dual-band cubic MIMO antenna demonstrates excellent performance, making it a highly promising candidate in continuous, movable, and varying body posture scenarios for several wireless biomedical implants.

REFERENCES

- [1] S. Movassaghi, M. Abolhasan, J. Lipman, D. Smith, and A. Jamalipour, "Wireless body area networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 16, no. 3, pp. 1658–1686, Third Quarter 2014.
- [2] S. Kiani, P. Rezaei, and M. Fakhr, "A CPW-fed wearable antenna at ISM band for biomedical and WBAN applications," *Wireless Networks*, vol. 27, no. 1, pp. 735–745, Jan. 2021.
- [3] G. Iddan, G. Meron, A. Glukhovsky, and P. Swain, "Wireless capsule endoscopy," *Nature*, vol. 405, May 2000, Art. no. 417.
- [4] M. Zada, I. A. Shah, A. Basir, and H. Yoo, "Ultra-compact implantable antenna with enhanced performance for leadless cardiac pacemaker system," *IEEE Trans. Antennas Propag.*, vol. 69, no. 2, pp. 1152–1157, Feb. 2021.
- [5] S. A. A. Shah and H. Yoo, "Scalp-implantable antenna systems for intracranial pressure monitoring," *IEEE Trans. Antennas Propag.*, vol. 66, no. 4, pp. 2170–2173, Apr. 2018.
- [6] A. R. Chishti et al., "Advances in antenna-based techniques for detection and monitoring of critical chronic diseases: A comprehensive review," *IEEE Access*, vol. 11, pp. 104463–104484, 2023.
- [7] A. S. Rao et al., "Concomitance of radio frequency energy harvesting and wearable devices: A review of rectenna designs," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 32, no. 12, Dec. 2022, Art. no. e23536.
- [8] A. Iqbal, M. Al-Hasan, I. B. Mabrouk, and T. A. Denidni, "Capsule endoscopic MIMO antenna with radiation pattern and polarization diversity," *IEEE Trans. Antennas Propag.*, vol. 71, no. 4, pp. 3146–3154, Apr. 2023.
- [9] A. J. Alazemi and A. Iqbal, "A high data rate implantable MIMO antenna for deep implanted biomedical devices," *IEEE Trans. Antennas Propag.*, vol. 70, no. 2, pp. 998–1007, Feb. 2022.
- [10] M. Khalid, M. Zahid, M. Yousaf, A. Akram, Y. Amin, and M. A. Qureshi, "An ultra miniaturized implantable antenna for gastrointestinal tract monitoring," in *Proc. IEEE Int. Conf. Emerg. Trends Eng., Sci. Technol.*, 2023, pp. 1–5.

- [11] Y. Fan, J. Huang, T. Chang, and X. Liu, "A miniaturized four-element MIMO antenna with EBG for implantable medical devices," *IEEE J. Electromagn., RF Microw. Med. Biol.*, vol. 2, no. 4, pp. 226–233, Dec. 2018.
- [12] A. Iqbal, M. Al-Hasan, I. B. Mabrouk, and M. Nedil, "Scalp-implantable MIMO antenna for high-data-rate head implants," *IEEE Antennas Wireless Propag. Lett.*, vol. 20, no. 12, pp. 2529–2533, Dec. 2021.
- [13] S. M. A. Shah, M. Zada, J. Nasir, O. Owais, A. Iqbal, and H. Yoo, "Miniaturized four-port MIMO implantable antenna for high-data-rate wireless-capsule-endoscopy applications," *IEEE Trans. Antennas Propag.*, vol. 71, no. 4, pp. 3123–3133, Apr. 2023.
- [14] A. Basir, M. Zada, Y. Cho, and H. Yoo, "A dual-circular-polarized endoscopic antenna with wideband characteristics and wireless biotelemetric link characterization," *IEEE Trans. Antennas Propag.*, vol. 68, no. 10, pp. 6953–6963, Oct. 2020.
- [15] R. Das and H. Yoo, "A multiband antenna associating wireless monitoring and nonleaky wireless power transfer system for biomedical implants," *IEEE Trans. Microw. Theory Techn.*, vol. 65, no. 7, pp. 2485–2495, Jul. 2017.
- [16] S. A. A. Shah, A. Basir, Y.-H. Lim, and H. Yoo, "A novel efficient wirelessly powered biotelemetric endovascular aortic stent antenna system," *IEEE Trans. Antennas Propag.*, vol. 71, no. 9, pp. 7132–7145, Sep. 2023.
- [17] A. Iqbal, M. Al-Hasan, I. B. Mabrouk, and M. Nedil, "A compact implantable MIMO antenna for high-data-rate biotelemetry applications," *IEEE Trans. Antennas Propag.*, vol. 70, no. 1, pp. 631–640, Jan. 2022.
- [18] S. N. Shah, M. Zada, J. Nasir, S. M. A. Shah, E. Arnieri, and H. Yoo, "Dual-band two-port MIMO antenna for biomedical deep tissue communication: Design, characterization, and performance analysis," *IEEE Access*, vol. 11, pp. 104622–104632, 2023.
- [19] D. Sharma, R. N. Tiwari, B. K. Kanaujia, S. Kumar, and K. Rambabu, "Dual-band MIMO antenna data telemetry for dual-chamber leadless cardiac pacing on Internet of Things environment," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 9072–9085, Mar. 2024.
- [20] M. Zada and H. Yoo, "A miniaturized triple-band implantable antenna system for bio-telemetry applications," *IEEE Trans. Antennas Propag.*, vol. 66, no. 12, pp. 7378–7382, Dec. 2018.
- [21] S. M. A. Shah, M. Zada, J. Nasir, O. Owais, and H. Yoo, "Ultraminia turized triband antenna with reduced SAR for skin and deep tissue implants," *IEEE Trans. Antennas Propag.*, vol. 70, no. 9, pp. 8518–8529, Sep. 2022.
- [22] S. A. A. Shah and H. Yoo, "Radiative near-field wireless power transfer to scalp-implantable biotelemetric device," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 7, pp. 2944–2953, Jul. 2020.
- [23] L. Chang, A. Iqbal, A. Basir, R. B. V. B. Simorangkir, and I. Ben Mabrouk, "Conformal MIMO ultrawideband antenna design for high-speed wireless telemetry in capsule endoscopy systems," *IEEE Trans. Antennas Propag.*, vol. 72, no. 7, pp. 5724–5732, Jul. 2024.
- [24] S. M. A. Shah, M. Zada, J. Nasir, O. Owais, and H. Yoo, "Four-port triple-band implantable MIMO antenna for reliable data telemetry in wireless capsule endoscopy and deep tissue applications," *IEEE Trans. Antennas Propag.*, vol. 72, no. 8, pp. 6229–6241, Aug. 2024.
- [25] R. Ali, Y. Cho, I. A. Shah, S. Hayat, A. Basir, and H. Yoo, "Compact dual-band MIMO implantable antenna system for high-data-rate cortical visual prostheses applications," *IEEE Trans. Antennas Propag.*, vol. 72, no. 8, pp. 6374–6386, Aug. 2024.
- [26] Q. Huang, H. Li, D. Jing, X. Ding, J. Jian, and S. Xiao, "A miniaturized triple-band MIMO antenna with dual-band circular polarization for leadless pacemaker," *IEEE Antennas Wireless Propag. Lett.*, vol. 23, no. 10, pp. 2835–2839, Oct. 2024.
- [27] F. Faisal, M. Zada, A. Basir, M. Chaker, and T. Djerafi, "Cost-effective 3D printable water-based MIMO and SISO antennas for high-data-rate biomedical implantable devices," *IEEE Trans. Antennas Propag.*, vol. 72, no. 8, pp. 6177–6186, Aug. 2024.
- [28] I. Nadeem and D. -Y. Choi, "Study on mutual coupling reduction technique for MIMO antennas," *IEEE Access*, vol. 7, pp. 563–586, 2019.
- [29] A. Iqbal, S. H. Kiani, M. Al-Hasan, I. B. Mabrouk, and T. A. Denidni, "A compact dual-band implantable MIMO antenna for wireless capsule endoscopy," *IEEE Trans. Antennas Propag.*, vol. 72, no. 10, pp. 7515–7524, Oct. 2024.
- [30] *IEEE Standard for Safety Levels With Respect to Human Exposure to Electric, Magnetic, and Electromagnetic Fields, 0Hz to 300GHz*, IEEE Standard C95.1-2019, Oct., 2019.
- [31] I. A. Shah, M. Zada, S. A. A. Shah, A. Basir, and H. Yoo, "Flexible metasurface-coupled efficient wireless power transfer system for implantable devices," *IEEE Trans. Microw. Theory Techn.*, vol. 72, no. 4, pp. 2534–2547, Apr. 2024.